

CF&SQL[ASSIGNMENT]

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Q1: Frog Jump

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] h = new int[n];
        for (int i = 0; i < n; i++) h[i] = sc.nextInt();
        int prev2 = 0;
        int prev1 = Math.abs(h[1] - h[0]);
        for (int i = 2; i < n; i++) {
            int one = prev1 + Math.abs(h[i] - h[i - 1]);
            int two = prev2 + Math.abs(h[i] - h[i - 2]);
            int curr = Math.min(one, two);
            prev2 = prev1;
            prev1 = curr;
        }
        System.out.println(prev1);
    }
}
```

Q2: Frog Jump with K Distances

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int k = sc.nextInt();
        int[] h = new int[n];
        for (int i = 0; i < n; i++) h[i] = sc.nextInt();
        int[] dp = new int[n];
        Arrays.fill(dp, Integer.MAX_VALUE);
        dp[0] = 0;
        for (int i = 1; i < n; i++) {
```

```

        for (int j = 1; j <= k && i - j >= 0; j++) {
            dp[i] = Math.min(dp[i], dp[i - j] + Math.abs(h[i] - h[i - j]));
        }
    }
    System.out.println(dp[n - 1]);
}
}

```

SQL JOIN Practice

1. INNER JOIN employees with departments

```

SELECT e.emp_id, e.emp_name, d.dept_name, d.location
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id;

```

2. LEFT JOIN all employees with department names

```

SELECT e.emp_id, e.emp_name, d.dept_name
FROM employees e
LEFT JOIN departments d
ON e.dept_id = d.dept_id;

```

3. Find departments with no employees

```

SELECT d.dept_id, d.dept_name, d.location
FROM departments d
LEFT JOIN employees e
ON d.dept_id = e.dept_id
WHERE e.emp_id IS NULL;

```

4. Employee name, project name, department name and hours
SELECT e.emp_name, p.project_name, d.dept_name, ep.hours
FROM employees e
INNER JOIN employee_project ep
ON e.emp_id = ep.emp_id
INNER JOIN projects p
ON ep.project_id = p.project_id
INNER JOIN departments d
ON p.dept_id = d.dept_id;

5. All projects and employees assigned to each project
SELECT p.project_id, p.project_name, e.emp_name
FROM projects p
LEFT JOIN employee_project ep
ON p.project_id = ep.project_id
LEFT JOIN employees e
ON ep.emp_id = e.emp_id;

6. Employee name and manager name using self join
SELECT e.emp_name AS employee_name, m.emp_name AS manager_name
FROM employees e
LEFT JOIN employees m
ON e.manager_id = m.emp_id;